Bayesian Linear Regression on the Communities and Crime Dataset

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Abstract

This project presents a complete Bayesian Linear Regression (BLR) pipeline implemented from scratch using the Communities and Crime dataset from the UCI Machine Learning Repository. The work includes data preprocessing, closed-form posterior computation, marginal likelihood evaluation, posterior predictive inference, and evidence maximization for hyperparameter tuning. The Bayesian model is compared against Ordinary Least Squares (OLS) and Ridge Regression in terms of predictive performance and uncertainty quantification. Additionally, a prior sensitivity analysis is conducted to study the influence of prior assumptions on posterior estimates and predictive behavior. The results show that Bayesian regularization improves generalization performance while simultaneously providing calibrated predictive uncertainty estimates.

1 Introduction

Regression analysis is one of the central tasks in statistical learning, where the objective is to model the relationship between explanatory variables and a continuous response variable. Classical methods such as Ordinary Least Squares (OLS) provide point estimates for the regression coefficients, but they do not naturally incorporate uncertainty in either the model parameters or the predictions.

Bayesian Linear Regression (BLR) addresses this limitation by introducing probability distributions over the regression coefficients. Instead of estimating a single optimal parameter vector, BLR computes a posterior distribution that combines prior assumptions with the observed data. Under Gaussian assumptions for both the likelihood and the prior distribution, the posterior admits a closed-form analytical solution, making BLR one of the most tractable Bayesian models.

An important advantage of the Bayesian framework is its ability to produce predictive uncertainty estimates through posterior predictive distributions. This allows the construction of credible intervals and uncertainty-aware predictions, which are particularly useful in real-world decision-making scenarios. Furthermore, the Bayesian framework provides the marginal likelihood, also known as the evidence, which can be used for principled hyperparameter optimization and model selection.

In this project, a complete Bayesian Linear Regression pipeline is implemented from scratch and applied to the Communities and Crime dataset from the UCI repository. The dataset contains socio-economic, demographic, and law-enforcement variables associated with violent crime rates across communities in the United States. Due to its relatively high dimensionality, substantial feature correlations, and missing-value structure, the dataset represents an appropriate benchmark for studying regularization, uncertainty quantification, and Bayesian inference in regression problems.

2 Objectives

The main objective of this project is to implement and analyze a complete Bayesian Linear Regression framework from first principles.

More specifically, the goals of this work are the following:

- To implement Bayesian Linear Regression from scratch without relying on probabilistic programming libraries.
- To derive and compute the closed-form posterior distribution of the regression coefficients.
- To compute posterior predictive distributions and construct 95% credible intervals for test predictions.
- To optimize the prior hyperparameters using evidence maximization through marginal likelihood optimization.
- To compare Bayesian Linear Regression against Ordinary Least Squares and Ridge Regression in terms of predictive performance and robustness.
- To identify and interpret the most influential predictors according to the posterior distributions of the regression coefficients.
- To analyze the sensitivity of the Bayesian model to different prior configurations and evaluate how prior assumptions affect predictive behavior and model conclusions.

3 Dataset and Preprocessing

3.1 Dataset Description

The analysis was conducted using the Communities and Crime dataset from the UCI Machine Learning Repository. This dataset combines socio-economic, demographic, and law-enforcement information collected from multiple public sources in the United States with the objective of explaining variations in violent crime rates across different communities.

The target variable considered in this work is `ViolentCrimesPerPop`, which represents the number of violent crimes per population unit. The predictor variables include information related to education, employment, income, housing conditions, racial distribution, urbanization, and policing.

The normalized version of the dataset was selected for the experiments because it simplifies numerical optimization and facilitates the interpretation of Bayesian priors.

Since the features are already scaled to comparable ranges, the regularization imposed by the Gaussian prior becomes easier to interpret and more numerically stable during optimization.

One of the principal challenges of the dataset is the presence of missing values, particularly in variables associated with law-enforcement statistics. Additionally, the dataset contains a relatively large number of correlated predictors, making it an appropriate benchmark for studying regularization and uncertainty quantification in regression models.

3.2 Preprocessing Pipeline

Several preprocessing operations were applied before fitting the regression models in order to ensure numerical stability, avoid data leakage, and improve the robustness of the analysis.

First, missing values encoded as `?` were converted into NaN values. Variables containing more than 30% missing values were removed from the analysis because excessive missingness could reduce the reliability of the corresponding predictors and introduce instability during model estimation.

For the remaining variables, missing entries were imputed using the mean value computed exclusively from the training set. This approach preserves consistency between training and testing conditions while avoiding leakage of information from the test data into the training process.

Non-informative attributes such as community names and geographical identifiers were also removed. These variables do not provide meaningful predictive information for regression and could potentially introduce artificial patterns unrelated to the underlying statistical relationships.

After cleaning the dataset, the observations were divided into training, validation, and test subsets. The training set was used for model fitting and hyperparameter optimization, while the test set was reserved exclusively for final evaluation.

Although the selected dataset version is already normalized, an additional standardization step was applied using statistics computed only from the training set. Features were transformed to approximately zero mean and unit variance in order to improve numerical conditioning and stabilize matrix operations during posterior computation.

Finally, an intercept term was added explicitly to the design matrix so that the Bayesian model could learn a global bias component independently from the explanatory variables.

The preprocessing pipeline was implemented carefully to avoid data leakage. In particular, imputation and standardization parameters were estimated exclusively from the training set and later applied unchanged to the validation and test subsets.

During implementation, generative AI tools were occasionally used as auxiliary support for debugging import issues, organizing the Python package structure, and refining portions of the preprocessing code. However, all preprocessing decisions, parameter choices, experimental configurations, and final validations were selected and interpreted manually by the authors.

4 Bayesian Linear Regression

4.1 Model Definition

Bayesian Linear Regression assumes the following linear relationship between the predictors and the response variable:

$$y = Xw + \epsilon$$

where:

- X is the design matrix,
- w is the vector of regression coefficients,
- ϵ represents Gaussian observation noise.

The likelihood function is defined as:

$$p(y \mid X, w, \beta) = \mathcal{N}(Xw, \beta^{-1}I)$$

where β denotes the noise precision parameter.
A Gaussian prior distribution is placed over the regression coefficients:

$$p(w \mid \alpha) = \mathcal{N}(0, \alpha^{-1}I)$$

where α represents the prior precision. This prior encourages coefficient shrinkage and acts as a probabilistic regularization mechanism.

4.2 Posterior Distribution

Under Gaussian assumptions for both the likelihood and the prior distribution, the posterior distribution over the regression coefficients also follows a Gaussian distribution:

$$p(w \mid X, y, \alpha, \beta) = \mathcal{N}(m_N, S_N)$$

where the posterior covariance matrix is given by:

$$S_N = (\alpha I + \beta X^T X)^{-1}$$

and the posterior mean vector is:

$$m_N = \beta S_N X^T y$$

The posterior mean represents the Bayesian estimate of the regression coefficients, while the posterior covariance matrix quantifies uncertainty in the parameter estimates.

The BLR model and posterior computations were implemented directly using NumPy linear algebra operations. Matrix inversion and posterior calculations were handled analytically without relying on probabilistic programming frameworks.

AI assistance was occasionally used to verify mathematical notation consistency and discuss numerical stability considerations for matrix inversion routines. Nevertheless, the derivation of the posterior equations, debugging of the implementation, and validation of the experimental outputs were carried out manually by the authors.

4.3 Posterior Predictive Distribution

For a new observation x_*, the posterior predictive distribution is defined as:

$$p(y_* \mid x_*, X, y) = \mathcal{N}(\mu_*, \sigma_*^2)$$

where the predictive mean is:

$$\mu_* = x_*^T m_N$$

and the predictive variance is:

$$\sigma_*^2 = \beta^{-1} + x_*^T S_N x_*$$

The predictive variance incorporates both observation noise and uncertainty in the regression parameters. This represents one of the principal advantages of Bayesian Linear Regression over standard frequentist approaches, since the model produces full predictive distributions instead of only point estimates.

4.4 Evidence Maximization

The hyperparameters α and β were optimized using evidence maximization through the MacKay fixed-point update procedure.

The update equations are defined as:

$$\alpha = \frac{\gamma}{m_N^T m_N}$$

$$\beta = \frac{N - \gamma}{\|y - X m_N\|^2}$$

where:

$$\gamma = \sum_i \frac{\beta \lambda_i}{\alpha + \beta \lambda_i}$$

and λ_i are the eigenvalues of $X^T X$.

The quantity γ can be interpreted as the effective number of parameters used by the model after regularization.

Starting from initial values $\alpha = 1.0$ and $\beta = 1.0$, the optimization converged after 17 iterations.

The final optimized hyperparameters were:

$$\alpha = 4862.36$$

$$\beta = 57.01$$

The relatively large value of α indicates that the evidence procedure favored substantial regularization. This result is consistent with the structure of the dataset, which contains many correlated predictors and potentially noisy explanatory variables.

During implementation, AI tools were also used as auxiliary support for discussing implementation alternatives and organizing portions of the optimization code. However, the final implementation, convergence analysis, interpretation of hyperparameters, and

all experimental decisions were validated and adapted manually according to the project requirements.

5 Bayesian Linear Regression

5.1 Model Definition

Bayesian Linear Regression assumes a linear relationship between the explanatory variables and the response variable:

$$y = Xw + \epsilon$$

where:

- X is the design matrix containing the predictor variables,
- w is the vector of regression coefficients,
- ϵ represents Gaussian observation noise.

The likelihood function is defined as:

$$p(y \mid X, w, \beta) = \mathcal{N}(Xw, \beta^{-1}I)$$

where β denotes the noise precision parameter.
A Gaussian prior distribution is placed over the regression coefficients:

$$p(w \mid \alpha) = \mathcal{N}(0, \alpha^{-1}I)$$

where α represents the prior precision. This prior acts as a probabilistic regularization mechanism by shrinking the coefficients toward zero.

5.2 Posterior Distribution

Under Gaussian assumptions for both the likelihood and the prior distribution, the posterior distribution also follows a Gaussian distribution:

$$p(w \mid X, y, \alpha, \beta) = \mathcal{N}(m_N, S_N)$$

where the posterior covariance matrix is given by:

$$S_N = (\alpha I + \beta X^T X)^{-1}$$

and the posterior mean vector is:

$$m_N = \beta S_N X^T y$$

The posterior mean m_N represents the Bayesian estimate of the regression coefficients, while the covariance matrix S_N quantifies uncertainty in the parameter estimates.

The Bayesian Linear Regression model and posterior computations were implemented directly using NumPy linear algebra operations without relying on probabilistic programming frameworks. Matrix inversions and posterior calculations were handled analytically according to the closed-form Bayesian solution.

5.3 Posterior Predictive Distribution

For a new observation x_*, the posterior predictive distribution is defined as:

$$p(y_* \mid x_*, X, y) = \mathcal{N}(\mu_*, \sigma_*^2)$$

where the predictive mean is:

$$\mu_* = x_*^T m_N$$

and the predictive variance is:

$$\sigma_*^2 = \beta^{-1} + x_*^T S_N x_*$$

The predictive variance incorporates both observation noise and uncertainty in the regression coefficients. This represents one of the main advantages of Bayesian Linear Regression over standard frequentist approaches, since the model produces full predictive distributions instead of only point estimates.

5.4 Evidence Maximization

The hyperparameters α and β were optimized through evidence maximization using the MacKay fixed-point update procedure.

The marginal likelihood, also referred to as the evidence, is defined as:

$$p(y \mid X, \alpha, \beta)$$

Maximizing this quantity allows the model to balance goodness of fit and model complexity in a principled Bayesian framework.

The optimization was performed iteratively using the following update equations:

$$\alpha = \frac{\gamma}{m_N^T m_N}$$

$$\beta = \frac{N - \gamma}{\|y - X m_N\|^2}$$

where:

$$\gamma = \sum_i \frac{\beta \lambda_i}{\alpha + \beta \lambda_i}$$

and λ_i are the eigenvalues of $X^T X$.

The quantity γ can be interpreted as the effective number of parameters retained by the model after regularization.

Starting from initial values $\alpha = 1.0$ and $\beta = 1.0$, the optimization converged after 17 iterations.

The final optimized hyperparameters were:

$$\alpha = 4862.36$$

$$\beta = 57.01$$

The relatively large value of α indicates that the evidence maximization procedure favored substantial regularization. This behavior is consistent with the structure of the dataset, which contains many correlated predictors and potentially noisy explanatory variables.

6 Experimental Results

6.1 Predictive Intervals

The posterior predictive distribution allows the construction of credible intervals for each prediction in the test set.

Examples of predictive intervals obtained from the Bayesian model are shown in Table 1.

Table 1: Examples of posterior predictive intervals

Prediction	Standard Deviation	Lower 95%	Upper 95%
0.1924	0.1356	-0.0733	0.4581
0.0004	0.1345	-0.2633	0.2640
0.1777	0.1349	-0.0868	0.4422

The predictive intervals quantify uncertainty by incorporating both observation noise and posterior uncertainty in the regression coefficients. The predictive standard deviation remained relatively stable across the test set, with an average value close to 0.135.

Some intervals include negative values despite the target variable being normalized between 0 and 1. This behavior is expected because Bayesian Linear Regression assumes Gaussian outputs and therefore does not explicitly constrain predictions to a bounded interval.

6.2 Selection of Influential Predictors

One of the principal advantages of Bayesian Linear Regression is that it provides full posterior distributions for the regression coefficients instead of only point estimates.

To identify the most influential predictors, the posterior mean coefficients obtained from the optimized Bayesian model were analyzed. Since all features were standardized during preprocessing, the magnitude of the posterior mean coefficients can be interpreted as a measure of relative importance across variables.

The predictors were ranked according to the absolute value of their posterior means:

$$|m_{N,j}|$$

where $m_{N,j}$ represents the posterior mean associated with coefficient j.

In addition to the posterior means, posterior standard deviations and 95% credible intervals were analyzed in order to evaluate the uncertainty associated with each predictor.

The variables identified as the most influential predictors are presented in Table 2.

Table 2: Most influential predictors according to posterior distributions

Feature	Posterior Mean	Posterior Std	95% Credible Interval
racepctblack	0.0427	0.0092	[0.0247, 0.0607]
PctIlleg	0.0367	0.0094	[0.0182, 0.0552]
NumStreet	0.0211	0.0055	[0.0104, 0.0318]
RentLowQ	-0.0196	0.0102	[-0.0396, 0.0004]
pctWInvInc	-0.0187	0.0096	[-0.0375, 0.0001]

Variables such as `racepctblack`, `PctIlleg`, and `NumStreet` maintained relatively stable positive coefficients with credible intervals that did not strongly overlap zero, suggesting stronger statistical support under the posterior distribution.

In contrast, variables such as `RentLowQ` and `pctWInvInc` exhibited intervals that nearly crossed zero, indicating weaker certainty regarding their contribution to the predictive model.

These relationships should be interpreted strictly as predictive associations rather than causal relationships.

6.3 Comparison with OLS and Ridge Regression

The Bayesian model was compared against Ordinary Least Squares and Ridge Regression using RMSE, MAE, and coefficient of determination (R^2) evaluated on the test set.

The comparison results are shown in Table 3.

Table 3: Comparison between regression models

Model	RMSE	MAE	R^2
OLS	0.1323	0.0940	0.6346
Ridge Regression	0.1309	0.0916	0.6423
Bayesian Linear Regression	0.1309	0.0916	0.6423

The results show that both Ridge Regression and Bayesian Linear Regression outperform Ordinary Least Squares across all evaluation metrics.

The weaker performance of OLS suggests that the Communities and Crime dataset contains substantial multicollinearity and noisy predictors. Since OLS does not include any regularization mechanism, the estimated coefficients become more sensitive to variance in the feature space, which negatively affects generalization performance.

Both Ridge Regression and Bayesian Linear Regression introduce regularization mechanisms that reduce coefficient variance and improve predictive robustness. Ridge Regression achieves this through an explicit L_2 penalty term, while Bayesian Linear Regression introduces regularization probabilistically through the Gaussian prior distribution imposed over the coefficients.

An important theoretical result is that the posterior mean of Bayesian Linear Regression under an isotropic Gaussian prior is closely related to the Ridge Regression solution. The experimental results empirically confirm this relationship, since both models achieved nearly identical predictive performance.

However, Bayesian Linear Regression additionally provides posterior uncertainty estimates and predictive distributions, which are not directly available in standard Ridge Regression.

6.4 Predictive Uncertainty Calibration

The Bayesian model achieved a 95% predictive interval coverage equal to:

$$0.9499$$

This indicates that approximately 95% of the observations in the test set fall inside the estimated credible intervals.

Therefore, the posterior predictive intervals appear to be well calibrated, suggesting that the Bayesian predictive distribution provides a coherent and reliable representation of uncertainty in the regression problem.

7 Prior Sensitivity Analysis

A prior sensitivity analysis was conducted in order to evaluate how different choices of the prior precision parameter α affect predictive performance, coefficient magnitudes, and posterior conclusions.

The analysis was performed by varying α across several orders of magnitude while keeping the noise precision parameter fixed. This experiment allows the study of how strongly the Gaussian prior regularizes the regression coefficients.

Three main regimes were observed during the analysis:

- For very small values of α, the Bayesian model behaves similarly to Ordinary Least Squares because the prior becomes weak and the likelihood dominates the posterior distribution.
- Moderate regularization improves predictive performance and stabilizes the regression coefficients.
- Extremely large values of α overshrink the coefficients toward zero and significantly reduce predictive accuracy.

For example, when:

$$\alpha = 10^{-4}$$

the model achieved:

$$RMSE = 0.1323$$

which is nearly identical to the result obtained using Ordinary Least Squares.

This behavior is theoretically expected because weak priors impose little regularization, allowing the Bayesian model to approach the classical maximum likelihood solution.

As α increased to intermediate values, predictive performance improved and the regression coefficients became more stable. This indicates that moderate regularization successfully reduces variance caused by multicollinearity and noisy predictors.

In contrast, extremely strong priors such as:

$$\alpha = 10^6$$

substantially reduced predictive performance:

$$RMSE = 0.1590$$

In this regime, the prior dominates the likelihood and excessively shrinks the coefficients toward zero, limiting the expressive capacity of the model and producing underfitting.

The coefficient norm also decreased considerably as α increased, clearly illustrating the shrinkage effect imposed by the Bayesian prior distribution.

An additional important observation is that several predictors remained influential across a broad range of prior configurations. Variables such as `racepctblack` and `PctIlleg` consistently appeared among the most relevant predictors even when the prior precision changed substantially. This suggests that some conclusions obtained from the model are relatively robust to prior assumptions.

Overall, the prior sensitivity analysis demonstrates that the Bayesian framework provides a flexible mechanism for controlling model complexity while simultaneously revealing how strongly prior assumptions influence posterior estimates and predictive behavior.

8 Conclusions

This project implemented a complete Bayesian Linear Regression pipeline from scratch using the Communities and Crime dataset from the UCI Machine Learning Repository.

The implemented framework included data preprocessing, closed-form posterior computation, posterior predictive inference, evidence maximization, model comparison, and prior sensitivity analysis. The entire Bayesian inference procedure was implemented analytically using linear algebra operations without relying on probabilistic programming libraries.

The experimental results demonstrated that Bayesian Linear Regression provides competitive predictive performance while additionally offering uncertainty quantification through posterior predictive distributions and credible intervals.

The comparison with Ordinary Least Squares and Ridge Regression showed that regularization substantially improves generalization performance on this dataset. In particular, both Ridge Regression and Bayesian Linear Regression achieved lower prediction errors than OLS, suggesting that the Communities and Crime dataset contains significant multicollinearity and noisy explanatory variables.

Furthermore, the strong empirical similarity between Ridge Regression and Bayesian Linear Regression confirmed the theoretical relationship between both methods under a Gaussian prior assumption. However, Bayesian Linear Regression additionally provides posterior uncertainty estimates and predictive distributions, representing an important advantage over standard deterministic regression approaches.

The prior sensitivity analysis highlighted the importance of selecting appropriate prior hyperparameters. Weak priors produced behavior similar to Ordinary Least Squares, while excessively strong priors degraded predictive performance due to overshrinkage of the regression coefficients.

Overall, the project illustrates the practical advantages of Bayesian methods in regression problems, particularly when uncertainty estimation, regularization, and interpretability are important considerations.

9 Reproducibility

All experiments were implemented in Python using NumPy, Pandas, scikit-learn, and Matplotlib.

The project structure was organized into modular preprocessing, modeling, and evaluation components in order to facilitate reproducibility and code readability.

Random seeds were fixed during preprocessing and dataset splitting to ensure consistent experimental results across executions.

During implementation, generative AI tools were occasionally used as auxiliary support for debugging, code organization, and refinement of certain implementation details. However, all methodological decisions, mathematical derivations, experimental configurations, validations, and result interpretations were carried out manually by the authors.

10 References

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